

V_{ud} extraction from EW corrections

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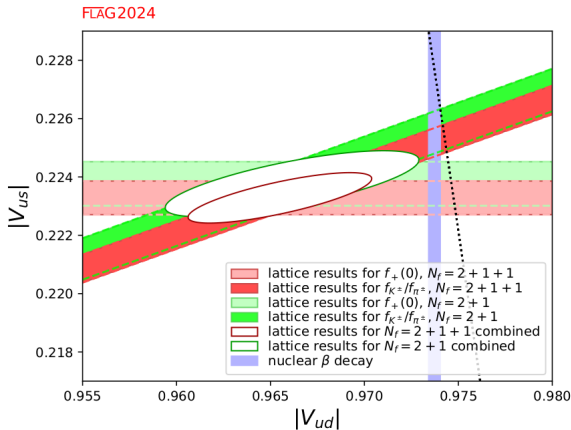
- ① CKM matrix
- ② V_{ud} from neutron decay
- ③ V_{ud} EFT framework
- ④ NLO EW corrections
- ⑤ NNLO EW corrections
- ⑥ SortBot Merging
- ⑦ SortBot Bottlenecks and changes
- ⑧ Changes in the SortBot Merge

- **CKM matrix** $\longrightarrow$ quark flavour mixing in weak interactions;
- First-row unitarity test

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1,$$

provides a sensitive probe of the SM;

- Current determinations of V_{ud} and V_{us} show a mild tension $\longrightarrow$ **Cabibbo Angle Anomaly**;
- Precision measurements of weak decays and accurate theoretical inputs are needed to clarify this tension;
- This talk:
 - Status of V_{ud} from superallowed β decays, pion decay and neutron decay;
 - Ongoing work on **NNLO EW corrections for neutron decay** and V_{ud} extraction.



V_{ud} and V_{us} using lattice QCD input and V_{ud} extracted from nuclear β decays, FLAG2024.

$$\frac{1}{\tau_n} = \frac{G_F^2 |V_{ud}|^2}{2\pi^3} m_e^5 f (1 + 3g_A^2) (1 + \Delta_R^V)$$

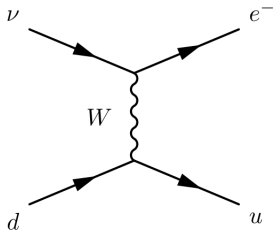
- f phase-space factor
- g_A axial coupling
- Δ_R^V universal electroweak radiative correction.

$$\Delta_R^V = 2.436(16)\%$$

$$|V_{ud}|_n = 0.97410(41)$$

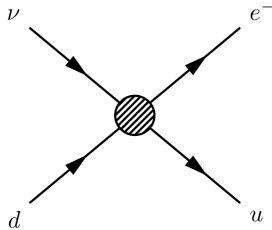
- First NLL calculation of mixed $O(\alpha\alpha_s^2)$ corrections;
- Reduced scale and scheme dependence;
- Improved CKM first-row unitarity tests;

$$d \rightarrow u e \bar{\nu}_e$$



Standard Model

$$m_W \gg k$$



Effective theory

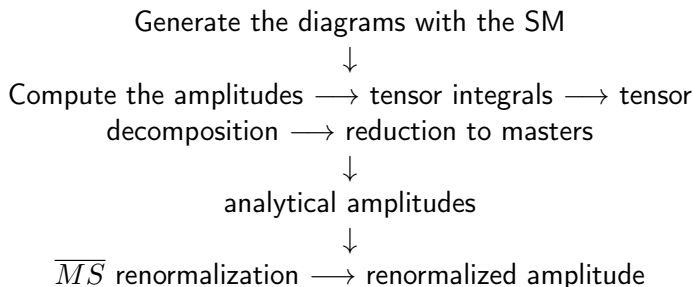
$$\mathcal{H}_{\text{eff}} = \frac{G_F}{\sqrt{2}} V_{ud} C_q(\mu) Q_q$$

$$Q_1 = (\bar{d}\gamma^\mu P_L u)(\bar{\nu}\gamma^\mu P_L e)$$

$$E_1 = (\bar{d}\gamma^\mu\gamma^\nu\gamma^\beta P_L u)(\bar{\nu}\gamma_\mu\gamma_\nu\gamma_\beta P_L e) - (16 - a_q^1\epsilon - a_q^2\epsilon^2)Q_1$$

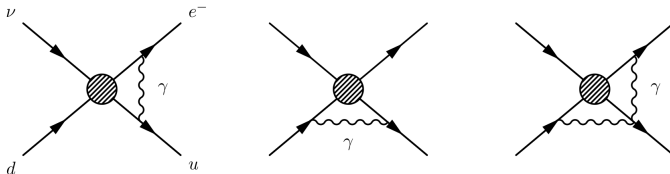
$$E_2 = (\bar{d}\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\gamma^\sigma P_L u)(\bar{\nu}\gamma_\mu\gamma_\nu\gamma_\alpha\gamma_\beta\gamma_\sigma P_L e) - (256 - b_q^1\epsilon - b_q^2\epsilon^2)Q_1 + c_q^0 E_1$$

- d dimensions (vanish in 4 dimensions);
- scheme dependent;



Choice of kinematics: external momenta set to zero and all masses below the W mass are set to zero. Only masses present in our calculation will be m_W , m_Z , m_H , m_t .

1 loop EFT contribution for pion decay.



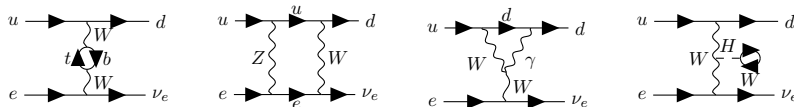
Exact decomposition of a propagator

$$\frac{1}{(k+p)^2 - m^2} = \frac{1}{k^2 - M^2} + \frac{-p^2 + 2k \cdot p - m^2 + M^2}{k^2 - M^2} \frac{1}{(k+p)^2 - m^2}$$

k and p stand for a linear combination of the internal moment and external momenta. The propagator mass is m and M is an introduced fictitious regulator.

Matching condition

$$\mathcal{A}_{\text{SM}} = \mathcal{A}_{\text{EFT}} = V_{ud} C_q Q_q$$



Representative one-loop EW diagrams contributing to the semi-leptonic process.

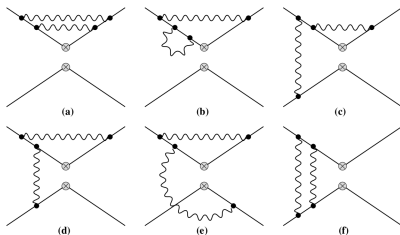
Wilson coefficient

$$C_q(\mu) = 1 + \frac{\alpha}{4\pi} C_q^{\text{NLO}} + \left(\frac{\alpha}{4\pi}\right)^2 C_q^{\text{NNLO}}$$

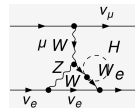
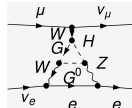
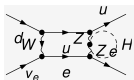
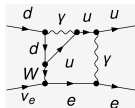
$$C_q^{\text{NLO}} = -5 + \frac{a_\mu^1}{4} + \frac{a_q^1}{12} - 2 \ln\left(\frac{\mu^2}{M_Z^2}\right)$$

- Matching performed at the electroweak scale.
- Universal corrections cancel in the ratio of the quark decay normalized to muon decay: $\frac{C_q}{C_\mu}$.
- Only box-diagram contributions survive.

NNLO EW corrections



EFT NNLO EW corrections.



SM NNLO EW corrections.

- NNLO requires inclusion of boxes and penguins
- Tadpoles are included in the matching: gauge independent NNLO EW corrections and potentially substantial contribution in scheme conversion: $\overline{MS}$ to On-Shell.

WC depends on μ :

$$\mu \frac{d}{d\mu} C_i^f(\mu) = \sum_j \gamma_{ij}^f C_j^f(\mu), \quad (1)$$

depends on active number of flavours;

Anomalous dimension for muon decay (physical operator)

$$\gamma_\mu^{(0)} = 0, \quad \gamma_\mu^{(1)} = \left(\frac{2}{3} N_\ell + \frac{8}{27} n_c N_u + \frac{2}{27} n_c N_d \right) (4 - a_\mu^1), \quad (2)$$

$$\gamma_q^{(0)} = -4, \quad \gamma_q^{(1)} = \left(\frac{16}{9} N_\ell + \frac{64}{81} n_c N_u + \frac{16}{81} n_c N_d \right) (a_q^1 - 8). \quad (3)$$

Solution of Renormalization Group Equations for the Wilson Coefficient

$$C^{(f)}(\mu) = J_f(\mu) U_f(\mu) U_f^{-1}(\mu_0) J_f^{-1}(\mu_0) C^{(f)}(\mu_0) \quad (4)$$

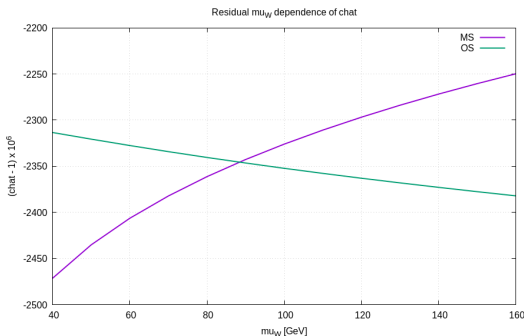
Evolution kernel at LO

$$U_f(\mu) = \left(\frac{\alpha^{(f)}(\mu)}{\alpha(M_Z)} \right)^{-\frac{\gamma^{(0)}}{2\beta^{(0)}}} \quad (5)$$

scheme independence

$$C = U_f^{-1}(\mu_0) J_f^{-1}(\mu_0) C^{(f)}(\mu_0) \quad (6)$$

Wilson-Coefficient residual scale dependence

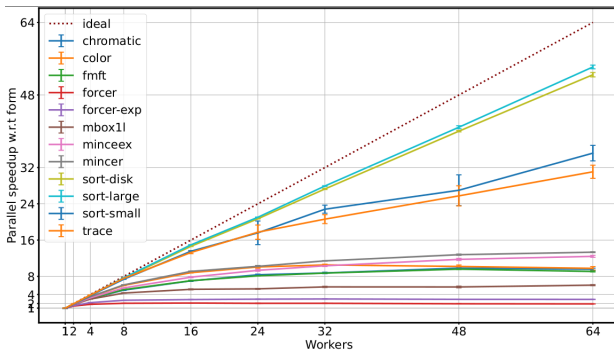


Scale dependence of the scale independent Wilson coefficient at NLO.

- Residual scale dependence at NLO induces an uncertainty $\delta C \simeq \pm 2 \times 10^{-4}$, which is significant for precision determinations of V_{ud} .
- NNLO corrections reduce the uncertainty to $\delta C < 2 \times 10^{-5}$, making perturbative uncertainties essentially negligible.

- Combination of lattice inputs and perturbative calculations;
- Higher order calculation + sub-percent lattice calculations;
- Current ongoing work on NNLO EW corrections for V_{ud} extraction from neutron decay.

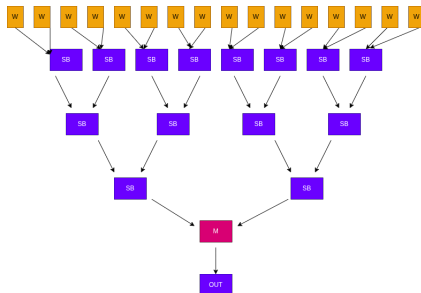
Parallel scaling at higher thread count



- sort-disk, sort-large, sort-small, trace scale up to 64;
- scaling is not good for other tests for > 16 workers.

SortBot Merging

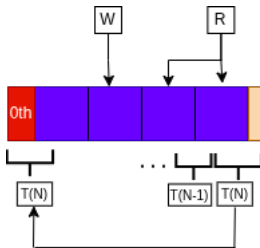
- **Master** $\rightarrow$ distributes terms to the workers;
- **Worker** $\rightarrow$ independent stream of sorted terms;
- N **workers** $\rightarrow N - 2$ **SortBots** are created;
- **SortBot** routine starts;



- **Master** performs the final merge and writes output.

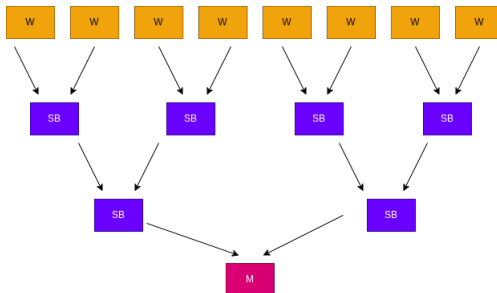
SortBot: Overlapping of terms

- SortBot $\longrightarrow$ ring buffer with blocks;

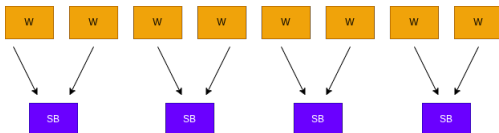


- Terms are allowed to overlap between blocks;
- 0th block** - the overlap in the end of term is copied to **0th block** so that the terms are contiguous.

- **Bottlenecks:** reading of the input with **division of terms** from **workers** to **SortBlocks**; inverse tree at the end with the writing to the file by the **Master**;

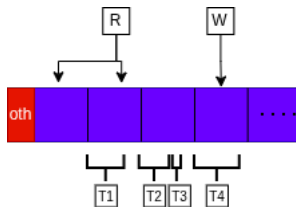


Bottlenecks

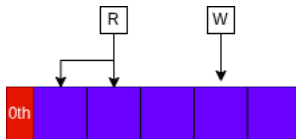


- Default buffer size increased over time → bigger **SortBlocks**;
- Result from **worker** fits into **single block** → SortBotTree mechanism does not parallelize
- If a **term** does not fit at the end of a block: Writer fills the **SortBlocks** all the way up to the end of the block, **overlapping** to the next;
- Once a block is completely full, it reaches the top and moves on to the next block.

Change: SortBot Block size

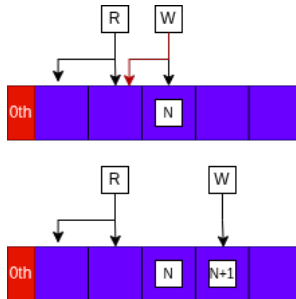


- Require fewer **SortBlocks** - master buffers not expanded so often;
- Whole terms must fit into the block: if it does not fit, it goes into the next block;



- Reading thread always locks pair of blocks - the current one and the previous;
- Writing thread locks the block where it is writing;

Change: SortBot Locking

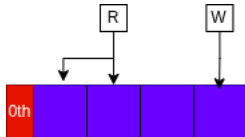


- Writer (W) detects if there is a Reader (R) waiting for it;
- W filling the blocks tries to obtain a lock of the block before the current fill block;
- If W does not obtain the lock, R thread already has the lock - very soon will be done and ready for more terms;
- The W moves to the next block;

- The number of blocks in each **SortBot** has to be at least 4 (due to the locking mechanism) - which corresponds to a minimum of 8 for **NUMBEROFBLOCKSINSORT**;
- There is a minimum number of terms that have to fit the block: **MINNUMBEROFTERMS**;
- Make sure a minimum number of terms is written: never make empty block - **MINWRITENNUMBEROFTERMS**;

PR #831

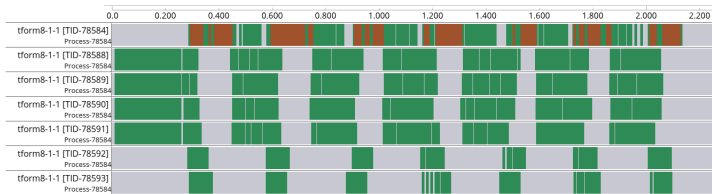
- **NUMBEROFBLOCKSINSORT** = 10 $\longrightarrow$ 8
- **MINNUMBEROFTERMS** = 10 $\longrightarrow$ 1
- **MINWRITENUMBEROFTERMS** = 1



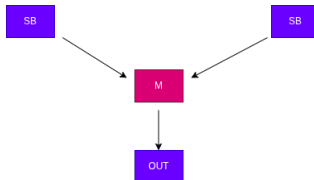
- No decrease in performance for any tests;
- Performance improvement for benchmarks where term merging is expensive (PolyRatFun, PolyRatFunExpand)
- Forcer, mbox1l

Results after changes: Profiler

- Profiler: AMDuProf with 4 workers;
- Test: generate a large number of terms, mostly sorting them;
- Green: Masters/threads in activity;
- Red: Master using **PutOut**;



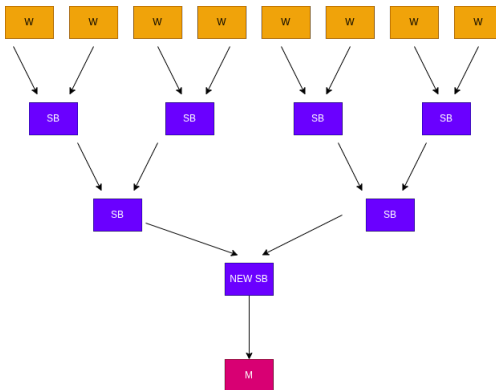
Bottlenecks



- End of the tree: **Master** performs last sorting as well as writing of terms into the output;
- Performance could be increased by having one last stage of sorting of the last 2 **SortBlocks** and **Master** only writing;

Future Work: Addition of new SortBot

- Add a new **SortBot** at the end of the merging tree to sort the streams that come from last 2 **SortBlocks**;
- **Master** only does the writing.



- Parallel performance at higher thread needs to be improved;
- **SortBot** changes - up to 30% improvement for some tests;
- Current working on improving **SortBot** parallelization by addition of extra **SortBot** at the end of the tree.